Thornier issues arise, though, when we think about the other kind of time travel, travel to the past. No doubt you are familiar with some of these. For example, there's the standard scenario in which you travel to the past and prevent your own birth. In many fictional descriptions this is achieved with violence; however, any less drastic but equally effective intervention—such as preventing your parents from meeting—would do just as well. The paradox is clear: if you were never born, how did you come to be, and, in particular, how did you travel to the past and keep your parents from meeting? To travel to the past and keep your parents apart, you had to have been born; but if you were born, traveled to the past, and kept your parents apart, you *wouldn't* have been born. We run headlong into a logical impasse.

A similar paradox, suggested by the Oxford philosopher Michael Dummett and highlighted by his colleague David Deutsch, teases the brain in a slightly different, perhaps even more baffling way. Here's one version. Imagine I build a time machine and travel ten years into the future. After a quick lunch at Tofu-4-U (the chain that overtook McDonald's after the great mad-cow pandemic put a dent in the public enthusiasm for cheeseburgers), I find the nearest Internet café and get online to see what advances have been made in string theory. And do I get a splendid surprise. I read that all open issues in string theory have been resolved. The theory has been completely worked out and successfully used to explain all known particle properties. Incontrovertible evidence for the extra dimensions has been found, and the theory's predictions of supersymmetric partner particles—their masses, electric charges, and so on— have just been confirmed, spot on, by the Large Hadron Collider. There is no longer any doubt: string theory is the unified theory of the universe.

When I dig a little deeper to see who is responsible for these great advances, I get an even bigger surprise. The breakthrough paper was written a year earlier by none other than Rita Greene. My mother. I'm shocked. No disrespect intended: my mother is a wonderful person, but she's not a scientist, can't understand why anybody would be a scientist, and, for example, read only a few pages of *The Elegant Universe* before putting it down, saying it gave her a headache. So how in the world could she have written *the* key paper in string theory? Well, I read her paper online, am blown away by the simple yet deeply insightful